Finite Element Modelling and Analysis of a Composite Rocket Body

MAE4426 Report 2

Athan Stefanos

33156476

Introduction

This report details the modelling, analysis, and evaluation of a real-world composite structural component using finite element analysis. Specifically, the effects of large holes on the stiffness and strength of a composite rocket body during flight. This report ties directly to the feasibility of side hatch-release parachutes for M-class [1] sounding rockets .

Component Choice and Problem Definition

The author of this report is a member of the Monash High Powered Rocketry (MHPR) team [2], which designs, validates, and manufactures competition rockets capable of reaching altitudes of up to 10 km and experiencing accelerations exceeding 20 g. Traditionally, recovery is achieved by separating the airframe into two sections to deploy parachutes. A more versatile approach is to employ a hatch that deploys parachute hardware without requiring airframe separation. However, introducing a cutout such as a hatch reduces the structural strength and stiffness of the rocket body. This report aims to investigate the reduction in strength that a side hatch hole in a composite body creates. Namely the reduction in bending stiffness and if a hatch will cause excessive deformation under launch loading.

Relevant Geometry and Simplifications

Sounding rockets designed for altitudes of 10–30 thousand feet commonly employ a 6-inch diameter body tube [2], which offers sufficient volume for onboard systems while maintaining a slender profile that minimises drag. Accordingly, a 6-inch diameter tube is selected for this study, with a total length of 2.5 m to align with previous competition rockets from MHPR. MHPR's latest rocket Zenith will be chosen to inform further choices for the geometry setup which can be seen in Figure 1.

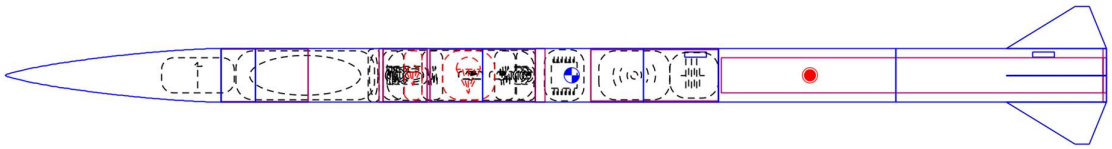


Figure 1: Zenith rocket configuration

Although rockets typically feature radial bolt holes, ports, and other cutouts, these are excluded bar the motor bulkhead to isolate and characterise the structural impact of a single large cutout. Couplers and bulkheads are likewise omitted, as they would alter the tube's strength characteristics. To reflect the hatch-based parachute deployment concept, a parachute bay is included in the design, since it directly interfaces with the cutout and must therefore be represented in the simulations.

The size of the parachute bay is determined by an ellipse cutting the surface of the body tube. To determine the size of this ellipse, it is obvious that the ellipse cannot equal nor exceed the diameter of the tube. Considering this a width of 130 mm will be set, equivalent to ~85% of the tube's diameter. Zenith's deployment bay had a volume of ~4 litres yet given the configuration not all this volume was required for storage of the main and drogue chute. For

lack of means to test the minimum required space for both parachutes, a reduction of ~38% will be applied to arrive at a final volume and ellipse size of 2.5 litres and 130 x 200 mm respectively [A1]. A diagram and dimensions of the setup can be seen in Figure 2.

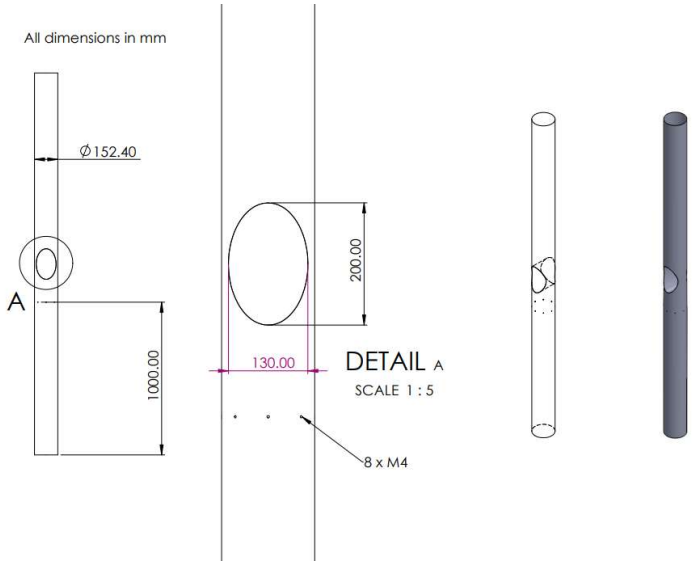


Figure 2: Geometry and diagram of test component

To model the interaction of the motor bulkhead with the tube, 8 x M4 radial bolt holes will be placed 1000 mm from the bottom of the body tube to simulate load paths of the acceleration through to the motor bulkhead. This utilises a flipped frame of reference that results in an identical load case, considering the bulkhead as stationary and the body providing downwards force. This can be seen further in the Boundary and Loading Conditions section.

Relevant Forces

During flight, a rocket is subjected to axial loads from thrust and aerodynamic drag. If these forces act perfectly along the rocket's longitudinal axis, the structure primarily experiences compression. In practice, however, small misalignments in the motor, perturbations in the airstream, or wind gusts introduce an angle of attack. This causes the aerodynamic forces to act off axis, generating bending moments about the centre of mass. The rocket's inertia provides the reactive force, resulting in structural bending loads in addition to the axial compression [3].

To find the accelerations and characteristics of the rockets flight path, OpenRocket was used with a detailed configuration of Zenith's flight profile to find the peak acceleration early in flight as seen in Figure 3. The rocket was simulated using the experimental thrust data from MHPR's student researched and designed hybrid rocket motor Solaris MkII, which has a total impulse of 9924 N/s and a peak thrust of 4 kN.

A sounding rocket normally experiences its peak acceleration early in its flight. Meaning the bending moments induced by an angle of attack would be less significant as the rocket is not travelling at a sufficient speed yet. The same can be said for aerodynamic drag. This relation

can be seen in Figure 3, showing a peak acceleration of 70.33 m/s^2 early in the flight. Hence, only axial loads from the rocket accelerating will be considered in this analysis.

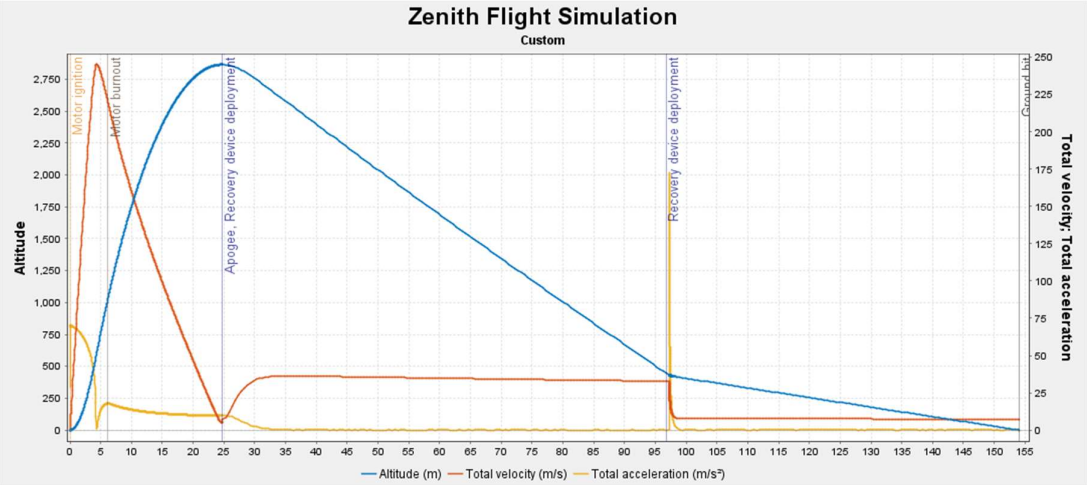


Figure 3: Zenith flight simulation of motion

Methodology – FEM Process

Boundary and Loading Conditions

Rockets normally utilise structural interfaces called ‘bulkheads’ that can be used for a variety of tasks such as housing components, hermetically sealing sections, and more relevantly, providing a means to transfer loads through to the airframe [4]. The thrust from the motor is normally transferred through a lip at the bottom of the airframe or through a bulkhead mounted at the top of the motor that interfaces with the body tube.

During acceleration, all components within the rocket body exert inertial forces into their respective bulkheads or attachment points. Each rocket can have a wide variety of layouts which cannot possibly be accounted for. For this reason, the average weight of Zenith will be applied as an axial force along the entire body tube. The bottom of the tube will be a tie constraint to emulate the fixture to the thrust applying motor.

Within Abaqus these were modelled as tie constraints and can be seen in Figure 4.

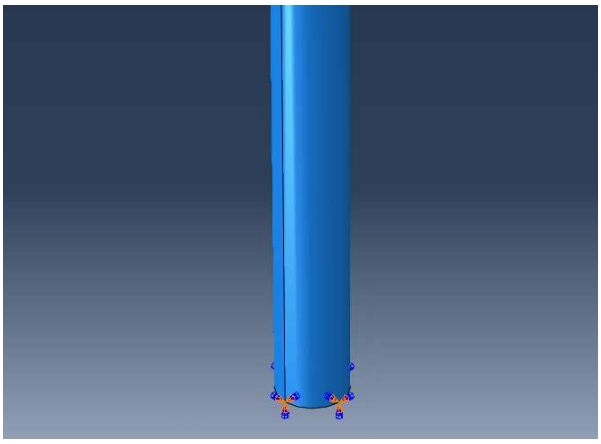


Figure 4: Bottom tube tie constraints

The bay was attached to the body tube through tie constraints with a hard contact interaction. This kept the interface between the bay opening and the tube opening together, simulating an epoxy bond to the body tube. The location and fixture of the bay is shown in Figure 5.

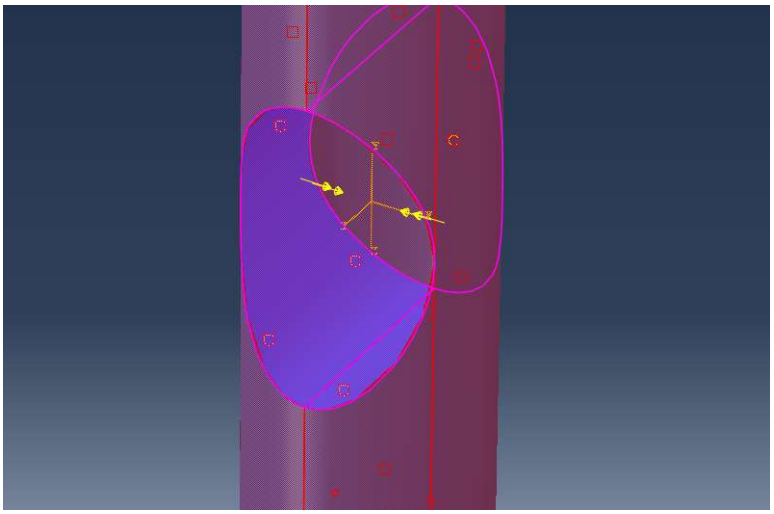


Figure 5: Bay-to-tube constraint and interaction

To simulate the launch loading a mass must be assigned to the entire airframe. Using the query tool, it was found that the body tube has a volume of 1,408,643 mm^3. The rocket Zenith has a wet weight of 31.8 kg [5], hence with some simple calculations it can be found that the density of the tube material must be 2.7 x 10^{-5} kg/mm^3. This assumes that the force of the rocket internal components is applied evenly along the entire airframe, which would change the specific load above and below the parachute hatch, yet since the motor makes up a large percentage of the mass it is deemed an adequate assumption, increasing apparent load for the simulation and hence FOS on simulation findings.

An acceleration of 70.33 m/s^2 was applied axially to simulate the launch loading early in the rockets flight. Combined with the artificially increased density of the tube this created an adequate representation of the launch loading that a mid-body hatch would experience. This was further confirmed by verifying the reaction forces equated to 2.679 kN. A diagram

summarising the loading can be seen in Figure 6, where the arrow ‘F’ represents the direction of acceleration.

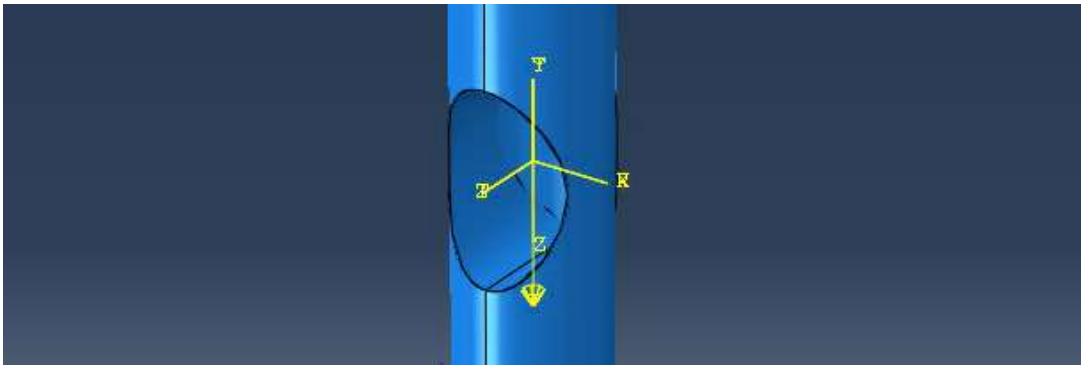


Figure 6: Acceleration direction diagram

Material Choice and Lay-Up Details

The material chosen for this report is fibreglass reinforced polymer. Fibreglass is a commonly used material in hobby rocketry and is favoured for its low weight, and high strength, while also retaining radio transparency. Fibreglass comes in a strand form. With thousands of strands making up ‘tows’ of fibreglass. These tows are then either made into unidirectional sheets or woven for multi-directional strength.

For this project a laminate will be chosen to replicate that of Zenith’s body tubes to most accurately gauge the effects for this format of rocket. This rocket has 6 layers of fibreglass utilising biaxial sleeve and 2x2 twill which resulted in a lamina thickness of 1.2 mm [5]. The lamina from the inside of the tube to the outside can be seen summarised in Table 1.

Table 1: Tube Lamina Summary

Number of Layers	**Layer (inner → outer)**
1	325 gsm -45/+45 biaxial sleeve
4	285 gsm S-Fibreglass 0/90 2 x 2 twill
1	325 gsm -45/+45 biaxial sleeve

Or more succinctly in lamina notation as $(\pm 45/(0/90)_2)_s$. To model this within Abaqus each ply will be broken down into individual layers so that they can be modelled off the properties of a single UD lamina.

P2362W-19L laminate from Toray Composite Materials America (TCMA) [6] will be used to model the laminate for its readily available mechanical testing data and its comparable weight and strength to the lamina used in project Zenith [5]. The data sourced in Table 2 is for testing in 22 degrees Celsius ambient conditions.

Table 2: Lamina mechanical properties

Property	Value (GPa)	
	Compressive	Tensile

E_1	130		148
E_2	10.1		8.5
G_{12}		3.94	
G_{23}		3.94	
v_{12}		0.30	
Longitudinal Strength	1779		3006
Transverse Strength	216		60.8
Longitudinal Shear		68.9	
Traverse Shear		96.5	

Meshing Approach

To begin meshing both the bay and the hatch S4R elements were used for the global mesh to capture the thin-walled rocket body well in low fidelity areas at a low computational cost. Quadratic interpolation elements were utilised in finer areas to better approximate bending effects with their non-linear interpolation. In addition, quad element shapes were used to mesh the cylinder for smooth edge flow and uniformity along the tube and around the hole to assist with mesh convergence.

Local element sizing was placed around the parachute opening for both the hatch and the tube in anticipation that this would be the area of the largest stress gradients. A biased sizing to the left and right sides of the hole were added to focus the mesh in these important areas which can be seen in Figure 7. This sizing began with a bias from 10 mm to 5 mm and was further decreased during mesh convergence.

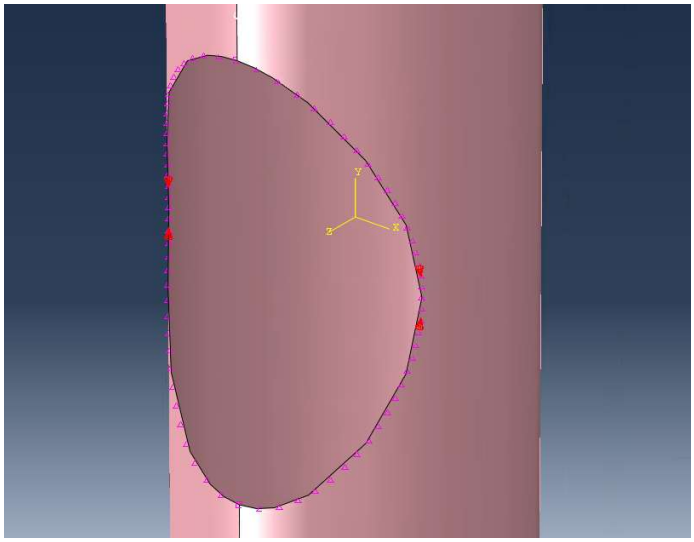


Figure 7: Hatch biased meshing

As detailed in results, stress concentrations were found around the sides of the holes during bending. To investigate the sensitivity of these stress concentrations with respect to the mesh, a mesh convergence was conducted.

Initially the bolt hole supports for the motor bulkhead caused large stress concentrations that fluctuated with mesh size, indicating no convergence. The bolt hole attachments were

deemed unnecessary with respect to the bending stiffness about the parachute hatch and were therefore excluded from the mesh. The following convergence study was conducted on the parachute and bay hatch interaction only.

The local mesh in the areas of the greatest stress was increased to monitor the evolution of stress with this refinement, with the aim of finding an invariance in stress results due to mesh size. The mesh was refined from 5 mm to 0.5 mm minimum sized elements and the stress results for this refinement can be seen in Figure 8.

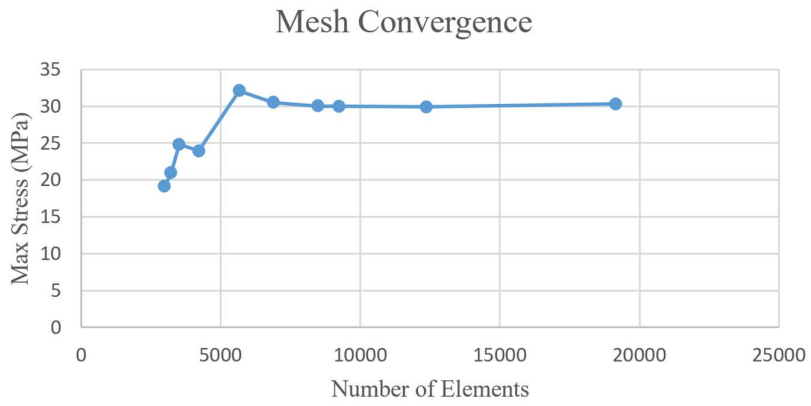


Figure 8: Mesh convergence plot

The stress plateaus with increase in mesh fineness indicating that the mesh is sufficiently detailed to capture accurate stress values. The mesh converged on the final stress value at ~6800 elements and showed no invariance up to ~19,000 elements. The final mesh around the hatch can be seen in Figure 9.

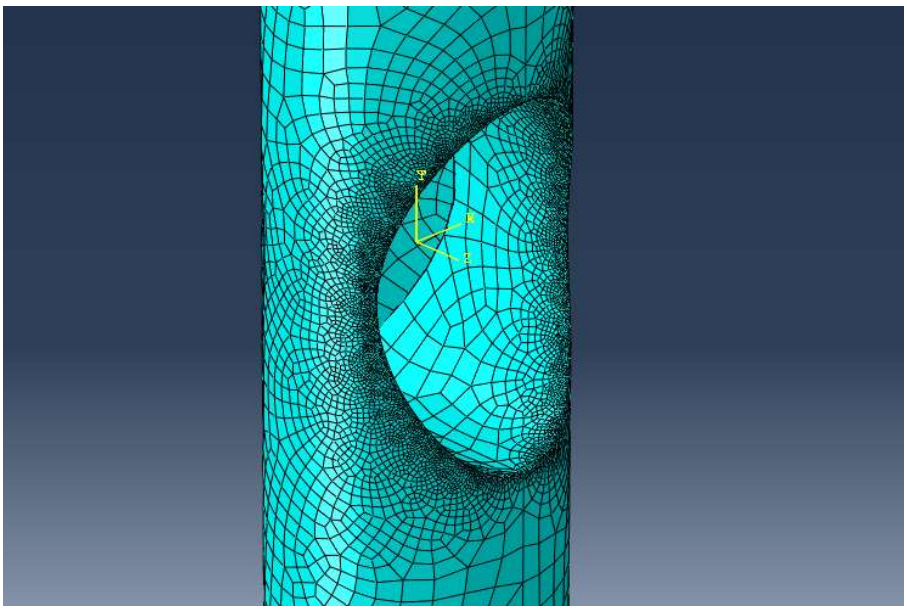


Figure 9: Final mesh around hatch

The mesh quality was high enough to converge and showed no adverse properties. More exactly, less than 0.05% deviated more than 80% from an ideal quad element. Furthermore, aspect ratio and skewness metrics for the final mesh can be seen in Table 3.

Table 3: Element quality metrics (aspect ratio, skew angles)

Element Metric	Minimum	Average	Maximum
Tube Element AR	N/A	1.49	4.31
Bay Element AR		1.64	5.1
Tube Element Angle	34	90.5	154
Bay Element Angle	30	91	148

Results

All simulation results within this report were found through Abaqus [7].

Stress & Strain

As anticipated the stress concentrations were around the edges of the hatch cutout. With the sides of the hatch buckling outwards and stretching the upper and lower sections of the hoop as seen in Figure 10. *Note the bay insert was included in the simulation but omitted visually for clarity.*

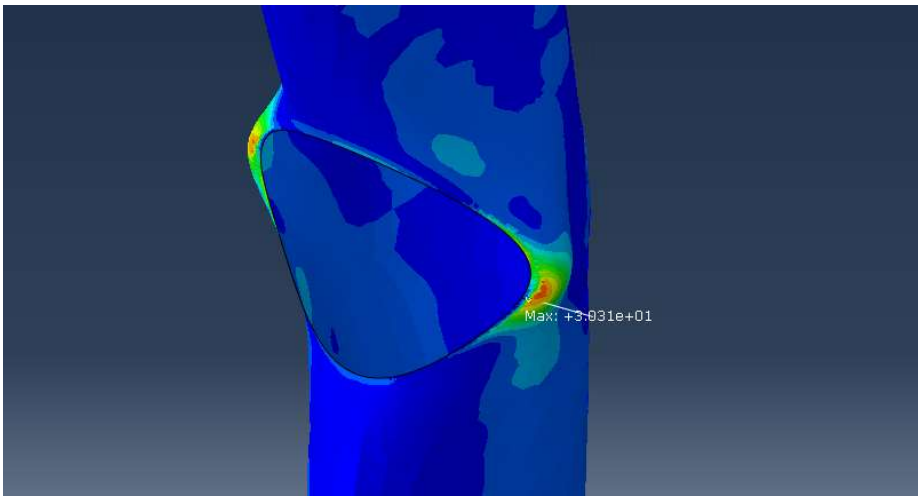


Figure 10: Von-Mises stress contours

A peak overall stress within the structure was found to be 30.313 MPa and occurred at the edges of the parachute bay hole slightly away from the edge. The stress contours for a selection of plies can be seen in Figure 11 highlighting the stress induced relative to the layer orientation. It can be seen on the 0º ply (defined axially along the cylinder) that most of the stress concentration occurs due to axial bending spanning the body tube. For the 45º layer the off axis bending caused at the edges of the hatch can be seen to induce the most stress. The 90º layer can be seen exhibiting the opposite behaviour to the 0º layer, exhibiting stress in deformation caused perpendicular to the body tubes axis.

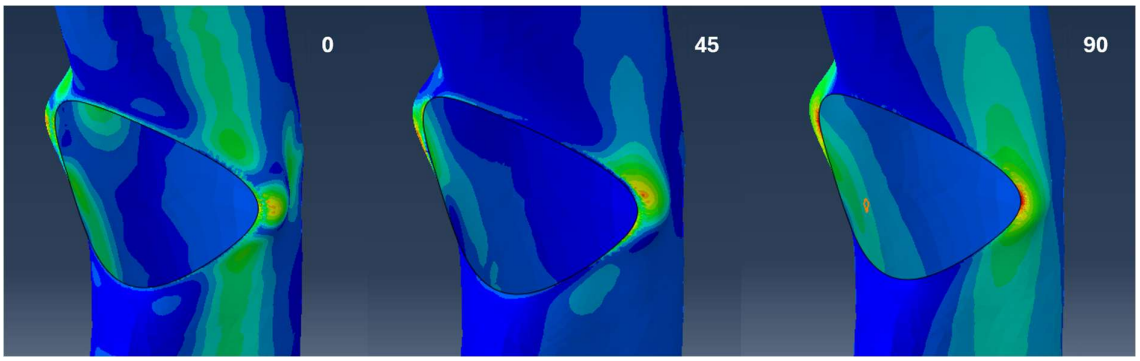


Figure 11: Specific layer stress contours

Due to the little deformation in the body tube the strain contours are quite homogenous as seen in Figure 12. The maximum strain occurred around the sides of the parachute hatch in compression as expected, where the highest stress concentrations were. There are also signs of slight stretching along the upper and lower edges of the bay.

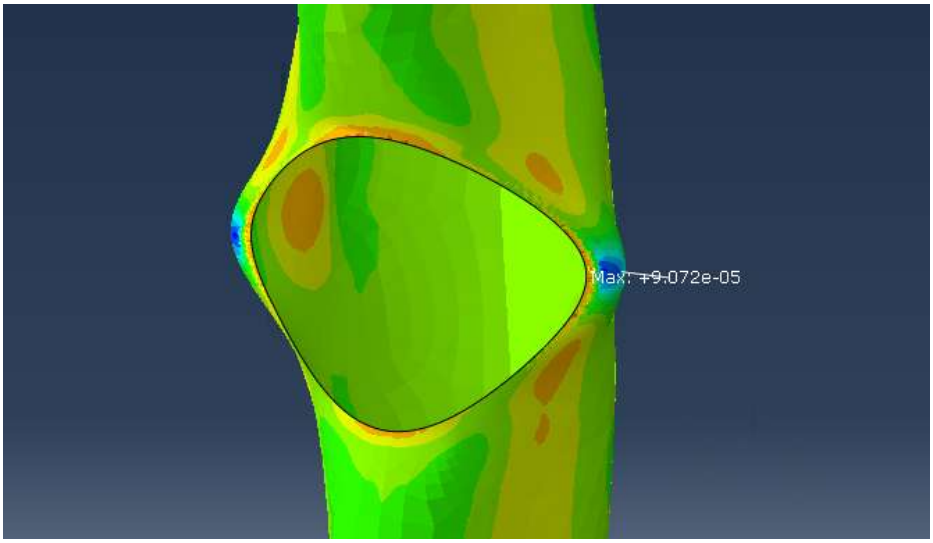


Figure 12: Strain contours

The maximum stress on the bay insert was lesser than the body tube at 24.183 MPa which occurred similarly on one of the lateral sides of the hole as seen in Figure 13.

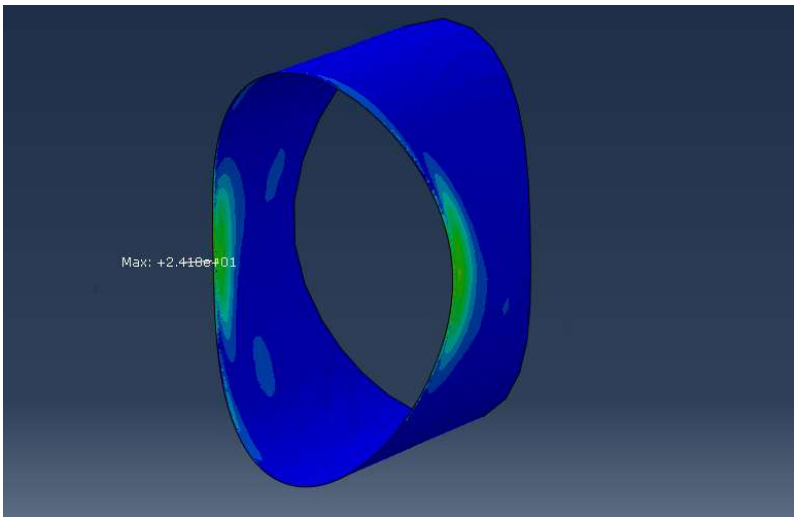


Figure 13: Bay insert stress contours

Deformation

The rocket body deformed a maximum of 0.4 mm at the upper end of the body tube. The major gradient of displacement occurred at the hatch where the stiffness of the tube was reduced. The deformed shape at a scale factor of 640 can be seen in Figure 14.

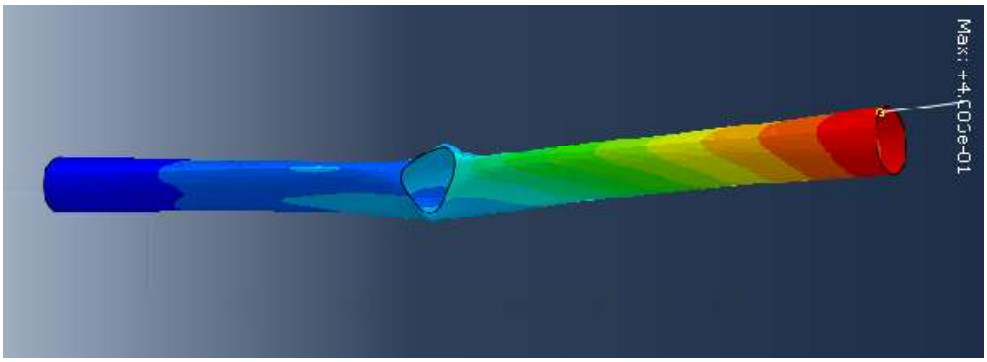


Figure 14: Body deformed shape (rotated)

Furthermore, the deformation contours can be seen in relation to the hatch more clearly in Figure 15.

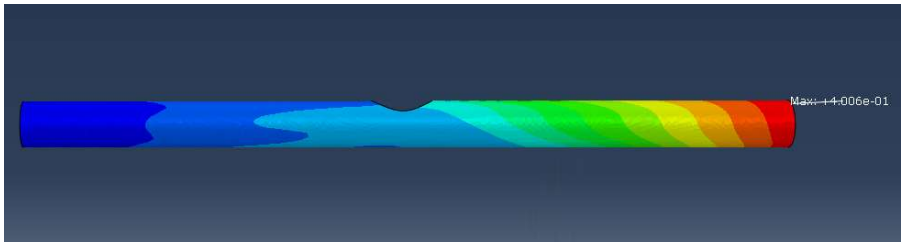


Figure 15: Deformation contours (side view)

Seen in Figure 16 the deformation gradient slightly decreases in proximity to the tube due to flexing and a ‘necking’ effect near the hole. This was likely caused by the addition of the

parachute bay insert providing additional lateral stiffness, causing the bending moment about the hole to cause this phenomenon around the hatch.

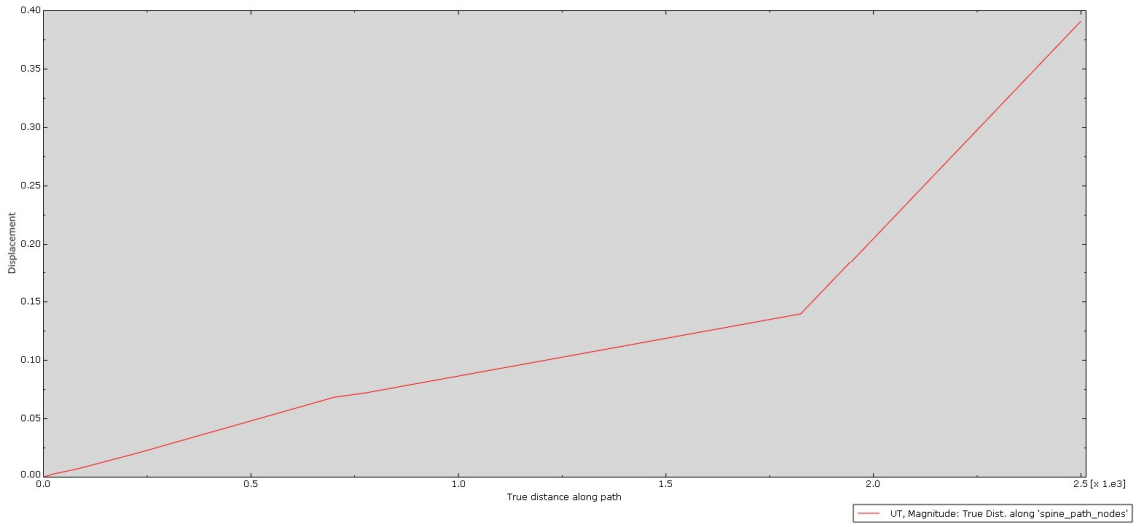


Figure 16: Deformation of tube along centreline

The curvature along the tube changed minimally as shown by the seemingly step increases in gradient, yet the mesh for this non-critical section was course and served only to demonstrate the overall deformation trend of the tube.

The bay insert deformed minimally with the top edge moving 0.12 mm as shown in Figure 17.

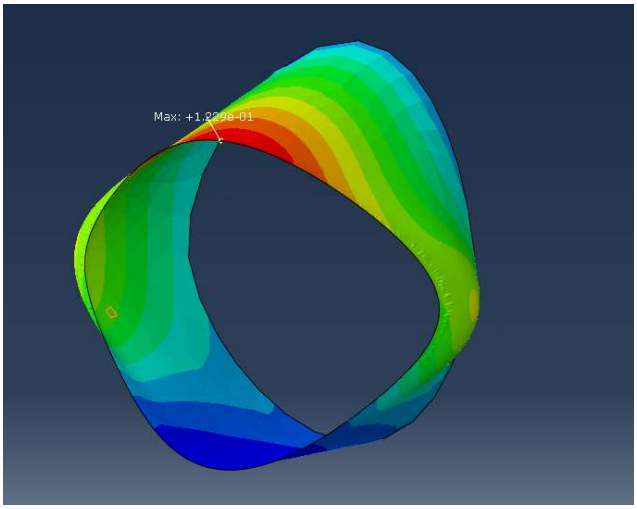


Figure 17: Bay insert deformation

Failure

To quantify failure the Hashin failure criteria [8] was simulated in Abaqus with the tensile, compressive and shear stresses in Table 2. This metric denotes failure when a scaled

interaction between these forces reaches a critical value of 1, indicating fibre failure. Both the compression and tension failure modes were not triggered, with the results shown in Figure 18 and Figure 19 for tension and compression respectively.

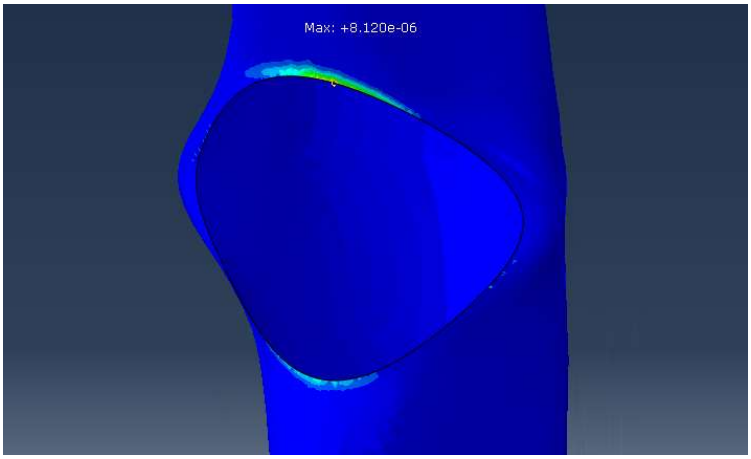


Figure 18: Hashin tension failure mode

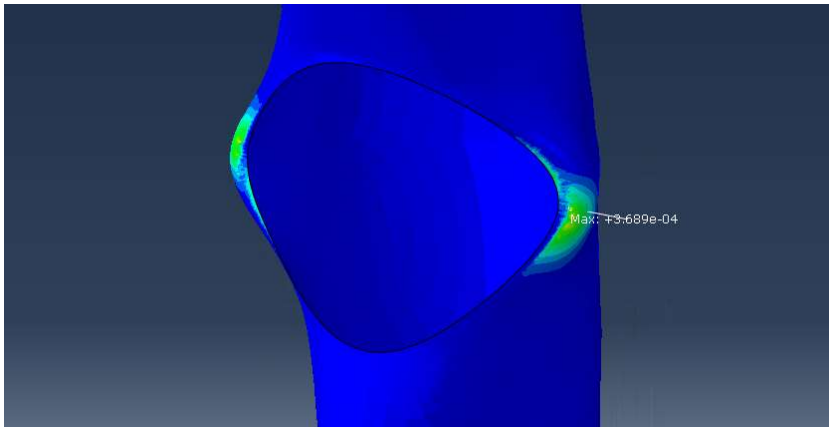


Figure 19: Hashin compression failure mode

Discussion

Stress and Strain

The simulation of the rocket body under axial loading remained within structural limits of the fibre reinforced polymer (FRP). Figure 10 clearly shows the stress concentrations occurring at the edges of the hatch due to buckling stress. This buckling stress presents strongly in the 90° plies of the laminate as seen in Figure 11 demonstrating the importance of diverse laminates in rocket bodies where even axial loads can result in shear stresses. The 45° ply stress contours from the same figure indicate that these diagonal plies play a major role in bending and shear stiffness, with the stress in these layers being concentrated around the areas of greatest strain.

The high axial compressive loads shown in the 0° ply that span the rocket body agree with literature indicating that largely offset plies more than 30 degrees from the 0° ply cause higher transverse tensile strains as seen in the 90° ply [9]. The stress of this load indicates that the hatch cutout will withstand launch loading. Outside of the initial axial loading early in flight, additional aerodynamic bending moments caused by wind gusts or an angle of attack can impose additional bending moments on this hatch. For this further reinforcement can be added, this could be additional plies of fibreglass to thicken the laminate around the stress concentrations or aluminium ring reinforcements. It should be noted that the additional attachment hardware for a hatch design should be adhered to the inner surface of the rocket body using an adhesive. Bolt holes introduced near the hatch cutout as noted in the mesh convergence study can introduce stress concentrations that can initiate failures in the laminate where other materials may interact with the composite structure [10].

The hatch insert reached a lower peak stress than the tube of 24 MPa. This insert added rigidity to the hole and prevented the tube from buckling along the central axis of the insert. The laminate thickness of this insert could be reduced by removing a pair of 0/90 plies to reduce weight.

Overall, if additional strength is desired the recommended improvement would be to implement additional plies local to the hatch oriented within 15 degrees of the 0° ply to better manage transverse tensile strains caused by the compressive buckling. This would reduce overall strain concentrations seen in Figure 12 by effectively changing the poisons ratio of the laminate for an axial compressive load [9].

Deformation

The maximum deformation at the top of the airframe was 0.4 mm as seen in Figure 14. This deformation would not adversely affect the angle of attack of the upper portion of the rocket but shows the trend of deformation caused by axial loading. In a higher loading case this deformation could increase and in turn increase aerodynamic bending moments caused by an angle of attack relative to the oncoming airstream. This in turn would increase deformation and form a positive feedback loop [11].

The axial loads within the flight regime of Zenith do not reach beyond the 2,679 kN after launch, not necessitating the need for reinforcement. Although if desired additional stringers could be attached along the length of the rocket to provide additional bending stiffness. Similarly to the reinforcement against stress and strain additional fibreglass layers could be added around the hatch to further limit tip deformation caused by bending about the hatch opening.

Failure

The two failure regions identified by the Hashin criteria are the upper, lower, and two lateral sides of the hatch cutout as seen in Figure 18, and Figure 19. The upper and lower failure mode is tension which agrees with the strain contours in Figure 12. This region is likely to depend highly on matrix shear stresses transferring this tensile load to the 90° fibres as the

circular cutout would mean that there are discontinuities in fibres if they are laid up regularly with the hatch being cutout afterwards.

The more important failure mode is the buckling on the sides of the hatch cutout. Thin-walled composites are especially prone to buckling which can significantly reduce the strength tolerance of structures below material indications [12]. This can cause matrix-fibre debonding and cause damage that can later propagate under a tensile force.

Considering this, a mitigation could be a specialised fibre laying process. For the tension failure mode, fibres could be curved along the upper and lower sections of the hatch so that there are continuous strands to better handle the tensile load after the hatch opening cut is made.

For the compression case additional plies could be laid locally to thicken the composite and better resist buckling loads. Additional aluminium plates could also be adhered on the inside of the tube to improve buckling stiffness and support the laminate in its through-thickness.

For the case of the Zenith fatigue failure is not a consideration considering the composites short lifespan of 3-5 launches. Despite this is fatigue failure were to initiate on the hatch it would likely be near the side areas where bending stresses were the highest.

Conclusion

The finite element analysis conducted in this report successfully quantified the structural effects of introducing a large side hatch into a composite rocket body. Under simulated launch acceleration conditions representative of MHPR's rocket Zenith, the FRP tube exhibited peak stresses of 30.3 MPa which is well within the allowable limits of the laminate. A maximum deformation of 0.4 mm was found indicating little change to the rockets shape during launch loading caused by bending. The Hashin failure criteria confirmed that neither fibre tension nor compression failure modes were triggered, indicating that the laminate layup and geometry provide sufficient safety margins for the expected loading during launch.

Analysis of the ply-level stress distributions revealed that the 0° plies primarily resisted the global axial compression, while the ±45° plies provided critical shear and bending stiffness around the hatch edges. The 90° plies experienced transverse tensile strains due to local compression and bending, highlighting the need for balanced laminates in thin-walled composite structures where even nominally axial loads can produce significant off-axis stresses.

Although the structure demonstrated adequate strength and stiffness, the regions adjacent to the hatch opening were identified as potential initiation sites for local buckling and matrix–fibre debonding. Reinforcement in these regions, either through the addition of local plies oriented within ±15° of the 0° direction or the use of bonded metallic stiffeners would further improve buckling resistance and reduce deformation. To address the Hashin tension failure mode, curved fibre layups around the perimeter of the hole could increase the structures resistance to tension stress in this localisation.

Overall, this report concludes that a side-hatch concept for parachute deployment in M-class rockets is structurally feasible for the format of 6-inch M-class sounding rockets. The finite element methodology and results for an axial load in this report can be referenced for the conceptualisation of parachute hatches in the future and used in conjunction with additional analysis to implement this concept within a sounding rocket.

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A. Appendices

A1. Bay Volume Code

```
en.diam = 25.4*6; % mm
ell.a = 130/2; % mm
ell.b = 200/2; % mm

fineness = 200;
[Xc, Yc, Zc] = cylinder(zen.diam/2, fineness);

surf(Xc, Yc, Zc*300 - 300/2)
hold on;

% Along the x axis there will be two rows, representing the start and end
% of the surface
Xe = [-zen.diam/2 *ones(1,fineness); zen.diam/2 * ones(1, fineness)]; % Start half diameter along X axis

% Y and Z should follow elliptical shape with same dimensional matrix

% Z direction is width Y Direction is height

theta = linspace(0,2*pi,fineness);

Ye = ell.a * cos(theta);  % ellipse along Z
Ze = ell.b * sin(theta);  % ellipse along Y


Ze = [Ze; Ze];
Ye = [Ye; Ye];

surf(Xe,Ye,Ze)
axis equal
```

```
%% Volume of intersection: ellipse ∩ cylinder
a = ell.a;
b = ell.b;
L = zen.diam;        % extrusion length (mm)
R = zen.diam/2;      % cylinder radius (mm)

% Numerical integration over Y–Z cross-section
N = 1200;                          % grid resolution
y = linspace(-b, b, N);
z = linspace(-a, a, N);
[YY, ZZ] = meshgrid(y, z);

ellipseMask = (YY.^2 / b^2 + ZZ.^2 / a^2) <= 1;
circleMask  = (YY.^2 + ZZ.^2) <= R^2;

dy = y(2)-y(1); dz = z(2)-z(1);

A_intersection = sum(ellipseMask(:) & circleMask(:)) * dy * dz;


V_intersection = A_intersection * L;

fprintf('Intersection area = %.2f mm^2\n', A_intersection);
fprintf('Intersection volume = %.2f mm^3 (%.4f L)\n', ...
        V_intersection, V_intersection/1e6);
```